

Rack A

Systems Specification

The engineering baseline for one fully specified rack: water, air, light, sensing and power, with the interfaces defined so a room of these scales without redesign. This document is the QC reference against which the built prototype is validated.

WATER	AIR	POWER	RACK SENSORS
14.5 L / tier	178 m ³ /h / tier	316 W / rack	2
BUILD COST	OPEN ITEMS		
₹51,256	2 · both tests		

Partly superseded. The drain-to-waste strategy in this document has been replaced by a closed-loop water recovery system — see *Closed-Loop Water Recovery* (RK-A-WRS Rev 2). The rack-side drainage hardware is unchanged; what changed is its destination, the fail-safe polarity of the diverter, and four control behaviours. Nineteen corrections from the integrated-model validation are recorded in the *Engineering Validation Record* (RK-A-QC Rev 1); the integrated rack envelope is now **1456 × 690 × 1960 mm at 125.4 kg dry**.

What this document changes

Specifying the subsystems properly rather than estimating them moved the cost a long way. That is the headline and it comes first.

LINE	PLACEHOLDER	ENGINEERED	WHY
Irrigation + drainage	₹3,500	₹8,950	Per-tier fill and drain valving, bulkheads, overflow standpipes, strainers, isolation
Electrical +	₹3,200	₹9,580	IP65 enclosure, isolator, RCBO, 24 V PSU, rack

control			controller, segregated cabling
Ventilation	—	₹5,520	Was never costed. Four EC fans and four perforated plenums
LED mounting	₹2,634	₹5,460	Hangers, saddles, IP65 connectors — rails alone were costed before
Rack sensors	—	₹1,650	Was never costed
Reflective panels	₹1,905	removed	Per your instruction. See the caveat below
Grow total	₹32,965	₹51,256	+55% — see §11 for the Rev 2 deltas

Correction — I double-counted, and you were right to push back. Rev 1 carried two separate parts doing one job: a folded GI plenum with a perforated face inside the ₹5,520 ventilation line, *and* a ₹600 PP "plenum face". The GI plenum already *is* the perforated duct that distributes air along the full 1176 mm. The PP panel was the old reflective rear panel repurposed, and it is genuinely redundant to the airflow function.

So the PP rear panel is now optional, as you asked, and the airflow system is unaffected by removing it. ₹600 comes out of the base rack. What the optional panel buys is tier *enclosure* — stopping air escaping sideways — plus the light recovery if specified in matte white.

What the optional panels still buy. Distribution along the tier is handled by the GI plenum and is not at risk. What is at risk without side and rear enclosure panels is *containment*: some supply air escapes sideways instead of crossing the canopy, so velocity uniformity will be worse than the ~10% coefficient of variation the CFD literature reports for a fully ducted tier. Measurable, not fatal — check canopy velocity at five points per tier at commissioning, and if the coefficient of variation exceeds 15%, fit the panels. The grid holes are already there.

Five rules everything else follows from

All water is in the rear service zone and below bed level. All electrical is above canopy or in the top-rear enclosure. Minimum 236 mm vertical separation, and no cable crosses a wet zone without a drip loop.

Flood depth is set mechanically

An overflow standpipe in each tray sets the level. A stuck-open fill valve overflows to drain rather than over the rim. The timer sets duration; the standpipe sets depth.

The grid is the only mounting interface

Every bracket, clip, sensor, plenum and rail bolts to the 50 mm hole grid. Nothing is drilled, welded or bonded to the frame in the field.

Trays are filled from a nozzle above the rim, not through a bottom port. Back-siphonage from a tray into the supply is then impossible by geometry rather than by a valve that can fail.

Any rack can be isolated alone

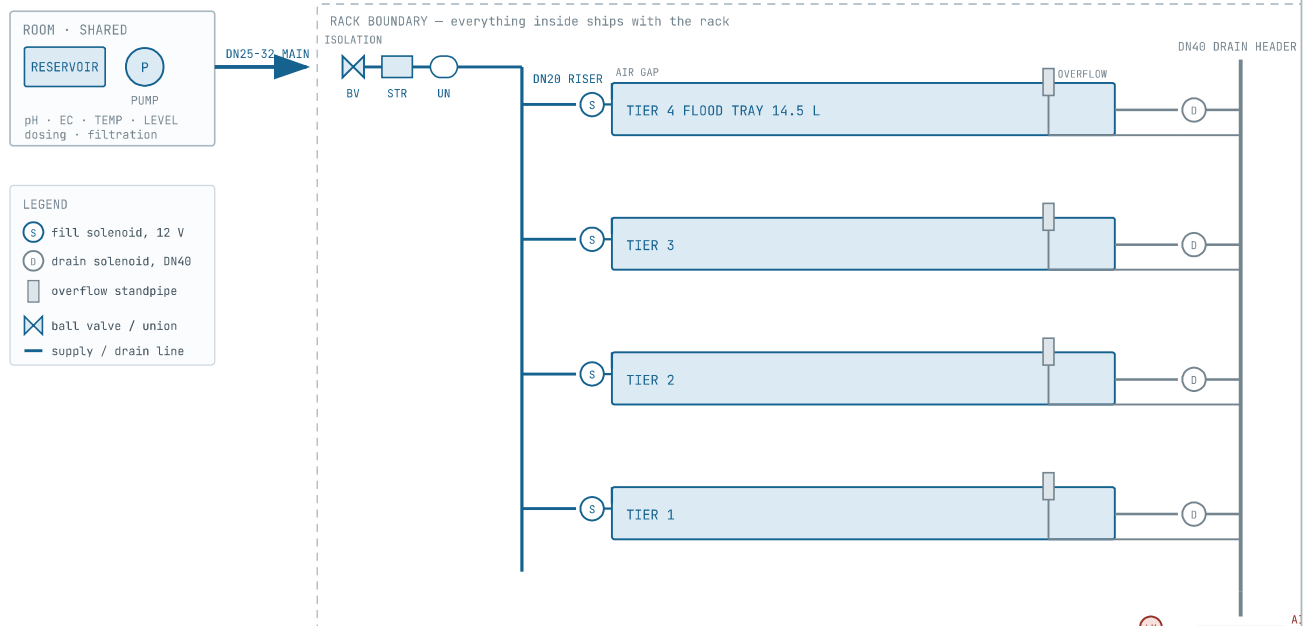
One ball valve and one union on water, one isolator and one RCBO on power, one addressable node on the bus. A rack comes out of service without stopping the room.

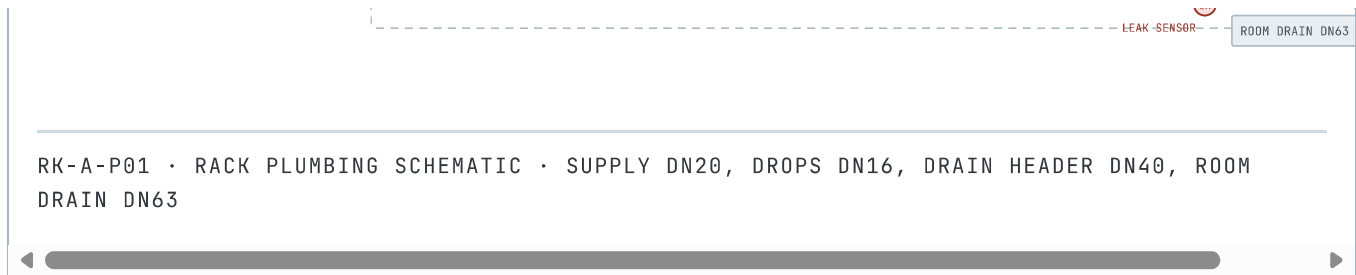
02 • IRRIGATION

Water into the rack

RACK PLUMBING SCHEMATIC

one rack • 4 tiers • fill from above with air gap • gravity drain





Connection to the rack, and where

POINT	LOCATION ON RACK	SIZE	FITTING
Supply inlet	Left rear upright, 250 mm above floor, rear service zone	DN20	PP ball valve + union, EPDM seat — the rack disconnects at the union
Inline strainer	Immediately after the inlet valve, serviceable from the aisle	DN20, 120 mesh	PP body, clear bowl, EPDM O-ring
Riser	Left rear upright, clipped to the 50 mm grid	DN20 PE	Push-fit, P-clips at 400 mm
Tier drop	Rear of each tier, terminating above the tray rim	DN16 PE	Push-fit elbow, nozzle clipped over the rim
Tray drain	Right end of tray floor, low point	DN40	PP tank connector, EPDM washer on the wet face, removable mesh strainer
Tray overflow	Adjacent bulkhead, standpipe crest at 30 mm	DN40 boss, DN32 pipe	PP tank connector + loose standpipe, tethered
Drain header	Right rear upright, vertical	DN40 uPVC	Clamp couplings so the stack lifts out without dismantling the rack
Discharge	Base of the right rear upright, 150 mm above floor	DN40	Air gap over the room drain — never a direct connection

How water actually reaches a tray

The tier drop terminates in a nozzle clipped over the tray rim, discharging downward into open air. There is no bottom fill port, no seal below the water line on the supply side, and **a physical air gap at every one of the sixteen fill points**. Back-siphonage of nutrient solution into the supply main is prevented by geometry rather than by a check valve that can stick. It also deletes four seals per rack that would otherwise be candidates for leaking.

Why the flood depth is mechanical

Each tray carries an overflow standpipe with its crest at 30 mm — above the 22 mm working flood, below the 47 mm rim. The fill solenoid runs on a timer, but if it sticks open the tray simply overflows down the standpipe into the drain header. The timer sets duration; the standpipe sets depth. This is the single most important safety feature in the water system and it has no moving parts.

Standpipe sizing caught a near miss. A DN25 standpipe passes 7.2 L/min at 8 mm of head over the crest — exactly the fill rate, so it would run at the point of flooding with no margin. **DN32 minimum**, which passes 11.9 L/min, a 1.65× margin. Specify DN32 and do not let a supplier substitute DN25 because it fits the same boss.

Sequential filling, and why it matters at room scale

STRATEGY	PUMP DUTY	SUPPLY SIZE	CYCLE	VERDICT
Sequential — one tier at a time	7.2 L/min	DN20 branch	24 min	Adopted
Simultaneous — all four tiers	29.0 L/min	DN25–32 branch	13.5 min	Rejected

Sequential filling costs eleven minutes of cycle time — irrelevant at one to three cycles a day — and buys a quarter of the pump duty and a smaller pipe. The real payoff appears at room scale: because the room controller fires one tier at a time *across all racks*, **the pump and the main never grow with rack count**. Twenty racks need the same 7.2 L/min pump as one. That is the property that makes this scale cleanly.

Seals and wetted materials

APPLICATION	MATERIAL	REASONING
All gaskets, O-rings, valve diaphragms	EPDM, food grade (WRAS/FDA)	Excellent with water at pH 5.6–6.2, and — the deciding factor — compatible with both hypochlorite and peracetic acid, so the sanitiser decision cannot strand it
Not to be used	NBR / Buna-N	Poor ozone and UV resistance; degrades in a lit, humid room
Not required	Viton / FKM	Chemically excellent, several times the cost, no benefit at these chemistries

these chemistries		
Valve and fitting bodies	PP or PVC-U, food grade	No metal in the wetted path — avoids both corrosion and metal ion pickup into the nutrient
Supply pipe	PE (LDPE) DN20/DN16	Standard Indian drip-irrigation ecosystem, push-fit, cheap, food safe, tolerant of movement
Drain pipe	uPVC DN40 to IS 4985	Rigid, solvent-welded, stocked everywhere
Tank connectors	PP body, EPDM washer on the wet face	Gasket on the wet side is standard practice — the water pressure seats the seal rather than trying to lift it
Fasteners in the wet zone	SS304 with nylon isolating washer	See the galvanic note in §10 — this is a real finding, not boilerplate

Routing so it does not get in the way

- **All water lives in the rear 90 mm service zone.** The riser and header run on the rear faces of the left and right uprights, clipped to the grid, outside the tray envelope.
- Tier drops cross the rear plenum face through grommets at a fixed position, then terminate over the tray rim. They do not cross the canopy, the LED plane, or the front access opening.
- The drain header is on the opposite upright from the supply riser, so a leak at either is unambiguous.
- Nothing water-carrying passes above bed level except the tier drop, which is above the rim and downstream of the air gap.
- Trays lift out upward with no plumbing attached to them except the two bulkheads, which disconnect at the clamp coupling on the header.

Multi-rack: how it scales

QUESTION	ANSWER
Common irrigation system?	Yes. One reservoir, one pump, one dosing set per room. DN32 supply main run along the rear wall, DN63 drain main below it at 1:100 fall.
Independent valve per rack?	Yes — mandatory. A manual ball valve plus a union at every rack inlet. This is what allows a rack to be worked on while the room runs.
Independent flow	Not needed, and worth avoiding. Flow is set per tier by the solenoid open time

Independent flow control per rack?	Not needed, and worth avoiding. Flow is set per tier by the solenoid open time, which the controller already owns. A balancing valve per rack is one more thing to be set wrong and drift. Add a per-rack flow meter on the Pro build for diagnostics, not control.
Isolating one rack for maintenance	Close the branch ball valve, break the union, lift the drain header clamp, open the rack isolator, pull the RS485 drop. Four actions, no room shutdown.
What grows with rack count	Only the length of the two mains and the number of branches. Pump, reservoir and dosing are sized by <i>concurrent</i> demand, which sequencing holds at one tier.

03 • DRAINAGE

Water out of the rack

PARAMETER	VALUE	BASIS
Method	Gravity throughout the rack	No pump above floor level. The only pump in the system is on the supply side.
Drain rate, DN40 at 22 mm head	30.7 L/min	4.2× the fill rate
Drain rate at 30 mm (overflow)	35.9 L/min	5.0× fill
Tray empty time	≈ 55 s	Declining head
Automated?	Yes – one DN40 drain solenoid per tier	Closed during fill and dwell, opened to drain
Tray fall to drain	2–3 mm across the floor	Formed into the tray so no puddle is left behind
Strainer	Removable mesh over the drain boss	Keeps media and roots out of the header
Discharge	Air gap at 150 mm above floor	Never a direct connection to the room drain

Why the overflow is a separate bulkhead rather than a concentric standpipe. The classic ebb-and-flow tray uses one boss with an inner overflow tube inside an outer drain screen. That

fails here in the one case that matters: if the boss blocks with media, both the drain *and* the overflow are lost at the same instant. Two independent penetrations mean a blocked drain boss still has a working overflow path. A blockage in the shared header defeats both, and that residual case is covered by the leak sensor and the solenoid interlock, not by geometry — it is listed in the FMEA.

04 • SENSORS

What is measured, and where

The governing principle is that a sensor belongs at the level where the quantity actually varies. Putting a CO₂ sensor on every rack measures the same room sixteen times and costs sixteen times as much.

SENSOR	LEVEL	QTY	EXACT LOCATION	HEIGHT
Air temperature + RH	RACK	1	Front-right upright, on the grid, in a vented radiation shield, mid-depth of tier 3	1239 mm — tier 3 canopy
Leak detection	RACK	1	Floor spot sensor at the base of the drain header, right rear	5 mm above floor
Air temp + RH, per tier	TIER	4	<i>Pro / R&D build only.</i> Same mounting rule, one per tier	canopy of each tier
PPFD / PAR	PORTABLE	1 per site	Quantum sensor on a grid-mount jig. Five points per tier at commissioning and after any fixture height change	canopy level
Leaf temperature (true VPD)	TIER	opt.	<i>Pro only.</i> IR thermometer on the grid, aimed at canopy centre at 45°	300 mm above canopy
pH	WATER	1	Reservoir or inline on the pump return, accessible for calibration without tools	reservoir
EC / TDS	WATER	1	Alongside pH, same manifold	reservoir
Water temperature	WATER	1	In the reservoir, mid-depth	reservoir

Reservoir level	WATER	1	Ultrasonic above the surface, or a float switch pair for low and low-low	reservoir
Flow rate	WATER	1	Inline on the supply main after the pump. Per-rack flow is a Pro option for diagnostics	plant room
Supply pressure	WATER	1	After the pump, for dry-run and blockage protection	plant room
CO ₂ (NDIR)	ROOM	1-2	Mid-room at canopy height, away from doors and the HVAC discharge	1200 mm
Room air temp + RH	ROOM	2	One at HVAC supply, one at return — the pair gives you the room's actual duty	supply / return
Smoke / heat	ROOM	1	Ceiling, per the room's fire design	ceiling
Room flood	ROOM	1	Floor low point, near the drain	floor

Two sensors that people buy and should not. There is no such thing as a VPD sensor — VPD is *computed* from air temperature, relative humidity and leaf temperature. In the standard build it is calculated from the rack T/RH node with leaf assumed equal to air, which is adequate for control; true VPD needs the IR leaf sensor and belongs only on the R&D rack. And there is no per-tray water level sensor in this design at all: **the overflow standpipe sets level mechanically**, which deletes sixteen sensors per rack and replaces them with a piece of pipe that cannot drift, fail or need calibration.

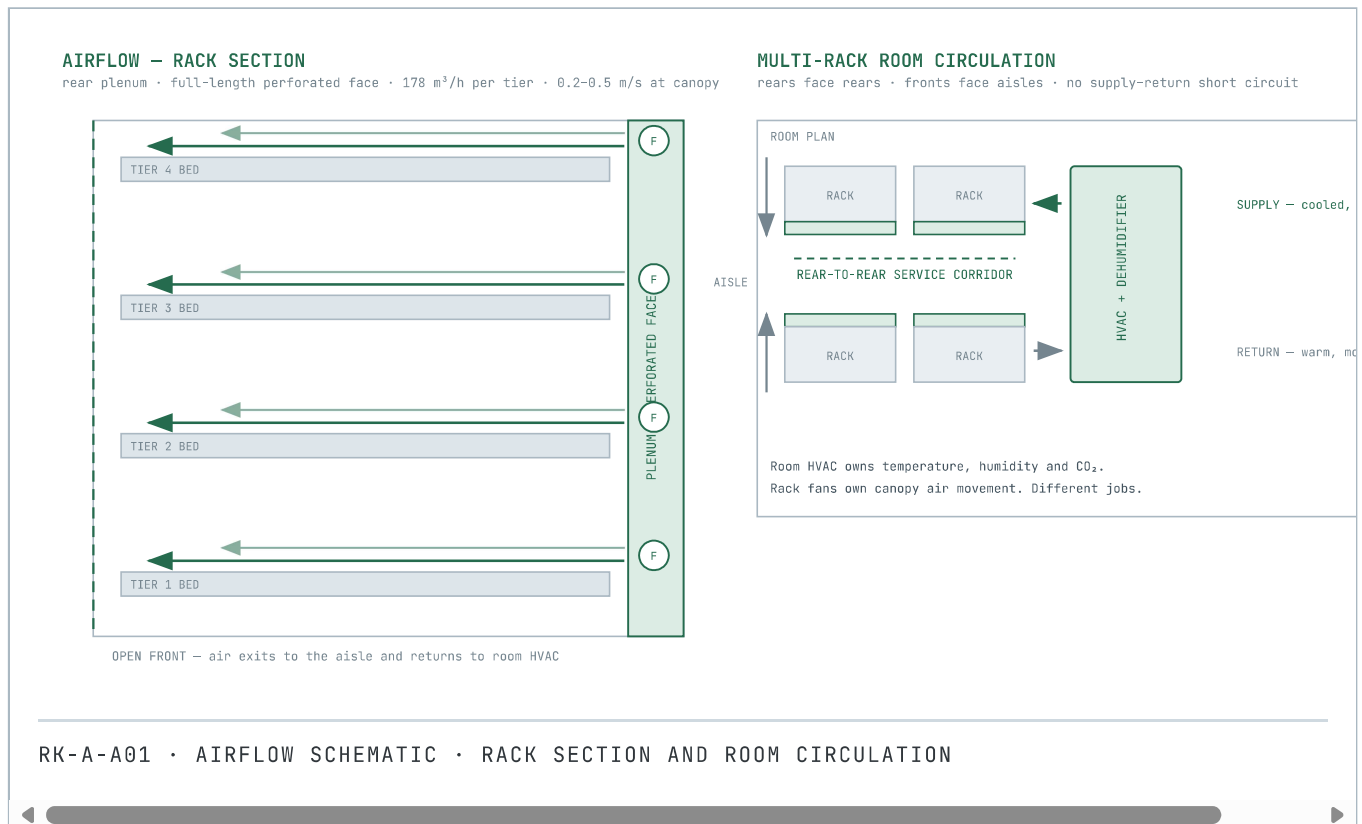
Mounting rules — how sensors avoid interference

- **Minimum 150 mm from any LED fixture.** Closer and the sensor reads fixture radiant heat, not air, and will report a temperature that no plant experiences.
- **Never in the direct plenum jet.** Mount at mid-depth of the tier, not against the perforated face — a sensor in the jet reads supply air, not canopy air.
- **Always in a vented radiation shield.** A bare sensor under horticultural lighting over-reads by a degree or more.
- **Above the tray rim, always.** No sensor may sit where a flood or an overflow can reach it.
- **Out of the tray path.** Nothing mounted where a tray is slid in and out — sensors go on the *front-right upright face*, clear of the 1176 mm tray opening.
- **Sensor cable in its own channel**, never bundled with mains or LED DC. Crossings at 90°.
- **Water chemistry sensors are never on the rack.** pH and EC probes need routine calibration

- **water chemistry sensors are never on the rack.** pH and EC probes need routine calibration and periodic replacement; putting them at the reservoir means one calibration for the whole room instead of one per rack.

05 • VENTILATION

Air through the rack



Rack fans or room HVAC — the answer is both, doing different jobs

These are not alternatives. **Room HVAC owns bulk temperature, humidity and CO₂. Rack fans own air movement at the canopy.** Room HVAC physically cannot deliver 0.2–0.5 m/s to sixteen canopies inside four enclosed tiers without ducting every tier of every rack and a fan far larger than the sum of the small ones — moving air 600 mm across a canopy takes about 5 W, whereas pushing conditioned room air through the same path takes an order of magnitude more.

For the initial Ooty stage, rack-level EC fans are both the efficient and the feasible choice: 20 W per rack, a bolt-on part with no room interface, and it lets you commission the racks before the room's HVAC is finalised. The room side then only has to hold air temperature and RH, which a conventional split unit plus dehumidifier does.

a conventional split unit plus dehumidifier does.

PARAMETER	SPECIFICATION	BASIS
Fan type	EC / DC axial, 120 × 38 mm	EC for speed control and low power; 38 mm depth for static pressure
Duty required	178 m ³ /h at ≈ 19 Pa	0.35 m/s over the 0.141 m ² canopy cross-section
Free-air rating to specify	≥ 254 m ³ /h	A fan on a perforated plenum delivers ~70% of free air
Bearings	Ball, not sleeve	Sleeve bearings fail early in high humidity and cannot run horizontally for long
Ingress	IP54 minimum	Condensing environment
Voltage / power	24 V DC, ≈ 5 W each	Same 24 V rail as solenoids and sensors
Mounting	Plenum end cap, right-hand end of each tier, anti-vibration grommets	Off the structure, so fan vibration does not enter the frame
Perforation	≈ 1590 × Ø4 mm at 16 mm pitch, 200 cm ² open	Face velocity 2.5 m/s, perforation loss 9 Pa
Direction	Rear plenum → forward across the canopy → open front → aisle → room return	Horizontal, entering along the full 1176 mm
Control	PWM or 0–10 V from the rack controller, with tachometer feedback	Tachometer is how a failed fan raises an alarm before the crop notices

Preventing stagnation and microclimates

- **Distributed inlet, not a point source.** This is the whole reason for the plenum: supplying along the full shelf length rather than from one end takes the velocity coefficient of variation from roughly 33% to about 10% in the published CFD work. A clip-on fan at one end of a shelf is the arrangement that evidence specifically rejects.
- **Fans run continuously**, including through the dark period. Stagnation at night is when condensation forms on leaves and disease starts.
- **Every tier has its own fan.** Tier-to-tier microclimates are prevented by giving each tier its own supply rather than hoping air finds its way

own supply rather than hoping air finds its way.

- **The alarm case matters more than the design case.** A tier whose fan has failed looks identical to a working one until the crop shows it. Tacho feedback and a controller alarm are not optional extras here.
- With side panels deleted, verify uniformity by measurement at commissioning — see §00.

Interaction with room HVAC and dehumidification. Every watt of LED energy that does not leave as photons leaves as heat, and all of the irrigation water that does not leave in the crop leaves as vapour. The rack fans do not remove either — they only move air across the canopy so the leaf boundary layer stays thin. The room HVAC and dehumidifier carry the whole thermal and moisture load. Sizing that is Phase 2 work; what the rack must do, and does, is present a predictable return air condition to the room rather than a set of stagnant pockets.

06 • LED MOUNTING

A mount, not an integration

The objective is that a fixture is a consumable and the rack is not. Everything below follows from that.

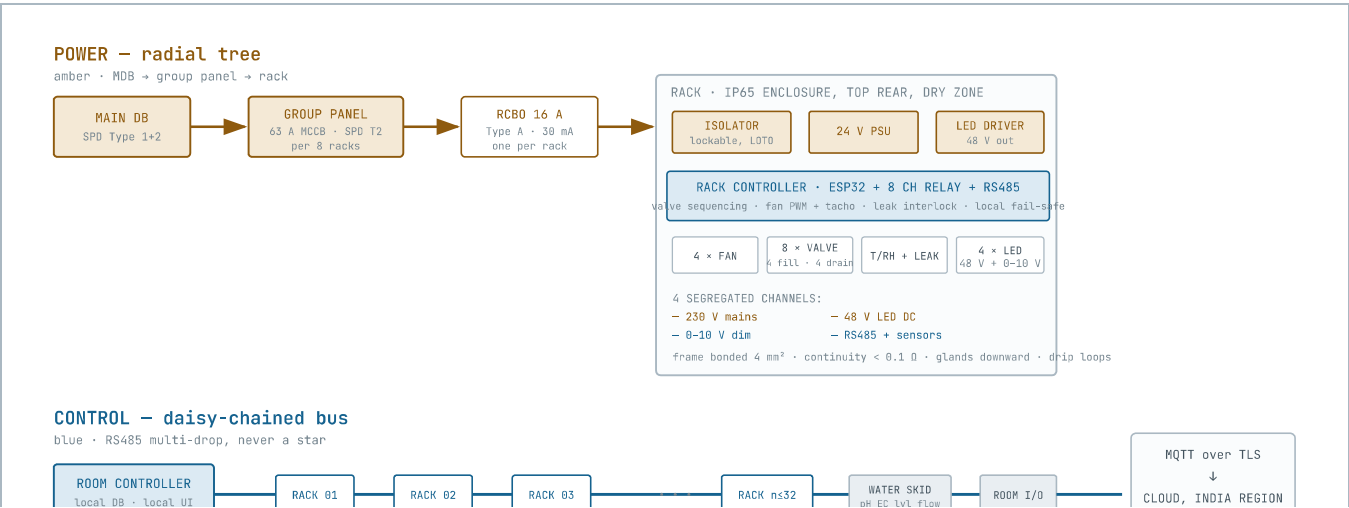
ELEMENT	SPECIFICATION
Rail	RK-A-301, aluminium 20 × 20 × 1.5, 1176 mm, two per tier at 30% and 70% of bed depth
Rail attachment	GI saddle bracket bolted to the 50 mm grid on the underside of the tier above, M8 SS304, nylon isolating washer between aluminium and galvanised steel
Height adjustment	Adjustable, continuously — M8 × 150 threaded-rod hangers through the rail with a nut above and a wingnut below, giving 0–100 mm of trim under a rail that sits on the discrete grid
Why threaded rod, not wire	Wire-rope hangers are faster but let fixtures swing, which makes PPFD uniformity unrepeatable and lets a fixture drift after someone brushes it while harvesting
Fixture interface	Two M8 clearance holes at 900 mm centres on the fixture top face. Product 1.2 conforms to this; any third-party bar that conforms will mount
Fixture envelope	1000–1200 mm long, ≤ 80 mm wide, ≤ 4 kg
Electrical	Fixture pigtail to an IP65 keyed connector at a fixed position per tier. A fixture change is

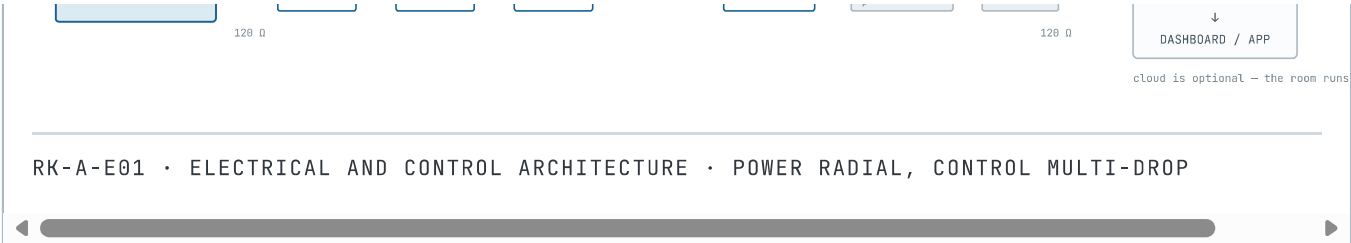
connection	never a rewiring job
Driver location	Remote , in the rack enclosure — not on the fixture. Keeps mass and heat out of the canopy, lets one driver serve a tier, and makes a driver failure a two-minute swap
Supply to fixture	Constant-voltage 48 V DC, or constant-current with output ≤ 60 V. The tier stays extra-low-voltage — no mains-voltage conductor enters the wet zone
Strain relief	P-clips to the rail at 300 mm centres, plus a 150 mm service loop at the fixture end so a fixture can be lowered for inspection without unplugging
Removal	Undo two wingnuts, unplug one connector, lift out. No tools beyond a spanner, no dismantling of frame, deck or plumbing
Future upgrade path	The interface is three things: the rail, the 900 mm bolt centres, and the connector. Any future fixture meeting those bolts on — including a higher-output bar for leafy greens, which the 80–200 mm clearance band already accommodates

The ELV decision is the one that matters for safety. Putting the driver in the rack enclosure and running 48 V DC to the fixture means the only mains conductors on the rack are inside an IP65 box mounted above canopy height. Everything at or below bed level — where the water is — is 24 V or 48 V. That, rather than any ingress rating, is what makes the electrical and water systems genuinely independent.

07 • ELECTRICAL AND CONTROL

Three levels





Your proposed topology, with one correction

You proposed Rack → Rack Controller → Group Control Panel → Main Distribution Board. That is right for **power**, and I would keep it exactly as stated. It is wrong for **control**, and the distinction is worth making because getting it backwards is a common and expensive mistake.

Power is a radial tree MDB → group panel (one per eight racks, 63 A) → per-rack 16 A RCBO → rack. Branches never rejoin. This is your architecture and it is correct.	Control is a multi-drop bus RS485 is a <i>bus</i>. Wiring it as a star through a group panel creates stub reflections and an unreliable link. The rack controllers daisy-chain along one cable, terminated 120 Ω at each physical end, with the room controller at one end.
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So the group panel is a power object, not a control object. Control skips it entirely and runs as one chain from the room controller through every rack.

Level 1 — the individual rack

ITEM	SPECIFICATION
Power input	230 V single phase, 1.37 A, via gland into the rack enclosure. Total connected load 316 W
Enclosure	IP65 polycarbonate 300 × 200 × 150, mounted top-rear on the services rail — above canopy, in the dry zone, glands facing downward
Local isolation	Lockable 16 A DP rotary isolator on the enclosure door, for lock-out/tag-out
Protection	16 A Type A RCBO, 30 mA, at the group panel — one per rack
Distribution	230 V → LED driver (48 V out) and 24 V DC PSU. 24 V → fans, solenoids, controller, sensors
LED connections	48 V DC + 0–10 V dimming pair per tier, IP65 keyed connector at the tier
Fan	24 V + PWM + tacho, 3-core to each plenum end cap

connections	
Valve connections	24 V switched by relay, eight circuits. Valves are the only electrical items below bed level, and they are 24 V
Sensor connections	I ² C to the T/RH node, dry contact from the leak sensor, in the dedicated channel
Controller	ESP32-class with 8-channel relay and RS485 transceiver. Runs the tier sequence locally and holds last-known-good setpoints, so the rack keeps irrigating if the bus or room controller drops
Emergency shut-off	Room-level E-stop, not rack-level. Latching mushroom head at each door, dropping the group contactor. A separate E-stop on each of sixteen racks is sixteen devices nobody can reach in an emergency; the rack isolator is for maintenance, which is a different job

Level 3 — sensors and communication

QUESTION	ANSWER	WHY
Sensor to controller	I ² C for digital T/RH (short run, inside the rack); dry contact for leak	No analogue runs longer than the rack
Wired or wireless	Wired	A room of galvanised racks is a poor 2.4 GHz environment, the cable route already exists for power, and Phase A calls for local-first reliability. Wireless stays available as a retrofit for a rack that cannot be cabled
Protocol	Modbus RTU over RS485 , shielded twisted pair, 120 Ω terminated both ends	Multi-drop to 32 nodes, 1200 m, immune to the electrical noise of LED drivers, and every PLC and gateway on the market speaks it
Controller to central	Daisy chain, one address per rack	Adding a rack is a spur and an address, not a new panel
Data collection	Room controller polls the bus, writes to a local time-series database	Local-first per Phase A §2.5 — the room keeps running and keeps logging with no internet
Cloud and app	MQTT over TLS to an India-region endpoint, dashboard and alerts on top	Optional layer. Nothing in the growing process depends on it

Level 4 — protection and safety

MEASURE	SPECIFICATION
Per rack	16 A Type A RCBO, 30 mA. Type A rather than AC because LED drivers and PWM fans produce pulsating DC residual currents that an AC-type device can miss
Per group	63 A MCCB, Type 2 SPD, one group per eight racks
Main	Type 1+2 SPD at the MDB, main earth bar
Earthing	TN-S per IS 3043. Every rack frame bonded with 4 mm ² green/yellow to the group earth bar. Continuity < 0.1 Ω, tested and recorded per rack
Water ingress	IP65 at and below bed level, IP54 above canopy. All glands face downward. A drip loop on every cable entering an enclosure
Cable management	Four segregated channels. Minimum 50 mm between mains and data, crossings at 90°
Emergency isolation	Room E-stop at each door, latching, dropping the group contactor. Rack isolator for LOTO
Interlocks	Leak sensor closes the rack's fill solenoids directly through the controller, and raises an alarm. This runs locally and does not depend on the bus

08 • BILL OF MATERIALS

Costed, panels removed

SUBSYSTEM	KEY CONTENT	COST
Core rack	Frame, decks, feet, anchors — unchanged from RK-A-MFG	₹12,391
Flood trays × 4	HDPE 3 mm food grade, white	₹4,835
Irrigation	4 fill + 4 drain solenoids, 8 bulkheads, 4 standpipes, strainers, isolation valve, PE pipework	₹7,650
Drainage	DN40 header, clamp couplings, air-gap discharge	₹1,300

Ventilation	4 EC fans, 4 perforated plenums, grommets, brackets	₹5,520
Plenum face OPTIONAL	Plain perforated white PP × 4 — enclosure and reflector only; the GI plenum already distributes the air	₹0 • + ₹600
LED mounting	8 rails, 16 hangers, 16 saddles, 4 IP65 connector pairs — <i>fixtures not included</i>	₹5,460
Electrical & control	IP65 enclosure, isolator, RCBO, 24 V PSU, rack controller, segregated cabling, bonding	₹9,580
Rack sensors	T/RH node in radiation shield, leak sensor	₹1,650
Powder coat	Optional on Grow	₹2,700
Rev 2 changes	Header level switch, moulded overflow collar, grouped fill manifold	+₹170
GROW — full automation		₹51,256
GROW-S — de-scoped	1 fill solenoid, single header drain valve, no rack controller, room T/RH only, mill GI	₹43,286

BUILD	COST	@ 40% GM	@ 45% GM	USE
Grow-S	₹43,286	₹72,150	₹78,700	Build this at Ooty. The room controller drives two racks directly — a per-rack controller earns its place at eight racks, not two
Grow	₹51,256	₹85,400	₹93,200	Commercial standard once rack count justifies distributed control

Read this honestly. A fully automated four-tier flood-and-drain rack with per-tier lighting, airflow and valving is a ₹43,000–51,000 object, and no amount of value engineering makes it a ₹10,000 one. The wire shelf it gets compared to is a shelf. This is a growing system with sixteen irrigation zones, four climate zones and a controller. **The comparison that matters is against importing an equivalent automated system, not against a shelf** — and that is the argument the product has to be sold on. If it must be sold against a shelf, the honest answer is to sell the Core rack at ₹20,650 and let the customer add subsystems as they grow, which is exactly what the modular architecture is for.

Where everything runs

ZONE	CONTENTS	RULE
Rear service zone, left upright	DN20 supply riser, DN16 tier drops	Water only. Clipped to the grid at 400 mm
Rear service zone, right upright	DN40 drain header, leak sensor cable	Water only, opposite side from supply so a leak is unambiguous
Rear plenum cavity	Fan cable, 24 V + PWM + tacho	Low voltage only, grommeted through the plenum wall
Top rear, above canopy	IP65 enclosure, mains, 48 V DC, RS485	Dry zone. Nothing at mains potential exists below this level
Front-right upright face	T/RH node, sensor cable	Clear of the 1176 mm tray opening
Under each tier	48 V + 0–10 V to the LED connector	In the LED rail P-clips, 150 mm service loop at the fixture
Front opening	Nothing	The full 1176 × 343 opening stays clear for tray handling

Failure modes and what they cost

Ranked by what actually goes wrong in wet, automated growing equipment rather than by what is easy to analyse.

#	FAILURE MODE	EFFECT	MITIGATION IN THE DESIGN	STATUS
F1	Tray drain bulkhead seal	Continuous leak onto the tier below, then	EPDM washer on the wet face, torque spec, 30-min flood test on	COVERED

	fails	the floor. Crop loss on two tiers	100% of trays, leak sensor and solenoid interlock	
F2	Fill solenoid sticks open	Would overflow the tray rim	Overflow standpipe at 30 mm — passive, no moving parts. Water goes to drain, not to the floor	COVERED
F3	Drain boss blocks with media or roots	Tray cannot empty	Removable mesh strainer, plus the overflow on an <i>independent</i> bulkhead so a blocked drain still has a path	COVERED
F4	Drain <i>header</i> blocks below both bosses	Both drain and overflow defeated. Tray fills to rim then spills	Rev 2: header high-level switch at 200 mm closes all fills before any tray is affected. Rodding eye at the base. See R1	COVERED
F5	Fan fails silently	Stagnant tier, condensation, tip-burn and disease. Looks identical to a working tier until the crop shows it	Tacho feedback per fan, controller alarm on loss of rotation	COVERED
F6	Galvanic corrosion, SS304 fastener in galvanised steel	Zinc is anodic to stainless and sacrifices at the washer face. White rust, then red rust, at every bolted joint in the wet zone	Nylon isolating washer under every SS fastener at or below bed level. Small stainless area against large zinc area keeps the rate low, but the washer face is where it shows	COVERED
F7	Galvanic corrosion, aluminium LED rail against galvanised saddle	Aluminium is anodic to zinc-coated steel in condensate. Pitting at the bracket	Nylon isolating washer at every rail-to-saddle joint. This is a new finding from this pass — it was not in the earlier BOM and is now a line item	COVERED
F8	Standpipe not refitted after cleaning	Tray will not hold a flood. Water runs straight through and the crop is not irrigated, silently	Rev 2: eliminated. The overflow is now a 30 mm collar moulded into the tray floor. Nothing to remove, nothing to lose. See R2	ELIMINATED
F9	Bus or room controller lost	Racks stop irrigating	Rack controller holds last-known-good schedule and runs locally. Loss	COVERED

			of bus is an alarm, not a stoppage	
F10	Condensate drips from fixture into tray	None — it is water going where water already goes	No action needed. Noted so nobody designs a drip tray for it	ACCEPTED
F11	Nutrient back-siphonage into supply main	Contamination of the whole room's water	Physical air gap at all 16 fill points. Prevented by geometry, not by a check valve	COVERED
F12	Residual current from LED drivers	An AC-type RCD may not trip on pulsating DC residual current	Type A RCBO specified, not Type AC	COVERED

Open items carried forward

ITEM	CLOSES BY
Depth-plane bracing — still open from the structural pass	Prototype sway test P3
Canopy velocity uniformity without side panels	Measure at commissioning, 5 points per tier

Manufacturing and maintenance review

Manufacturing Nothing added in this pass needs a process the shop does not already have. The plenum is a folded, perforated GI part — the same sheet operation as the deck. The bulkheads, valves and pipe are bought-out. The one new tolerance that matters is the tray drain and overflow boss positions, which must match the header spacing across all four trays.	Maintenance Trays lift out upward after releasing one clamp coupling. The drain stack lifts out as one piece. Fixtures come off with two wingnuts and a connector. Fans are in the plenum end cap, reachable from the aisle. The one poor-access item is the fill solenoid at the top tier, at 1825 mm — reachable, but a two-hand job at head height. Consider grouping all four fill solenoids on a single serviceable manifold at 1200 mm instead of one at each tier.
Water/electrical safety The strongest feature is that only 24 V exists at or below bed level. Mains is confined to one IP65 box above canopy. The weakest point is	Areas needing redesign Both closed in Rev 2. The fill solenoid position is resolved by the grouped manifold (R3) and the overflow route by the header level switch

IF65 box above canopy. The weakest point is not electrical at all — it is F4, a blocked header, where the only defence is detection.

the overflow route by the header level switch plus the moulded collar (R1, R2). What remains are two *tests*, not two design questions.

11 • RESOLUTIONS

Closing the open items

Three of the four are now closed by design change rather than by procedure. The fourth needs a test. One of these resolutions removes a failure mode entirely rather than mitigating it, which is always the better outcome.

R1·F4 — blocked drain header

The insight that resolves this is **where the water goes when the header blocks**. Trays drain into the header at their own tier heights, so a blockage below tier 1 causes the level to rise from the base and re-enter *tier 1's tray first*, through its own drain line. That makes the header itself the earliest and most certain place to detect the fault — well before any tray is at risk.

MEASURE	SPECIFICATION
Header high-level switch	Float or optical level switch in the DN40 header at 200 mm, just below tier 1's drain entry
Interlock action	Closes all four fill solenoids. Does not open the drains — draining into a blocked header makes the situation worse, not better. Raises an alarm
Wiring	Hard-wired to the rack controller as a local interlock, independent of the RS485 bus
Access	Rodding eye at the header base and a removable top cap, so the stack can be cleared without dismantling
Cost	+₹630

F4 moves from **RESIDUAL** to **COVERED**. The detection is upstream of the consequence rather than after it.

R2·F8 — standpipe left out after cleaning

Rev 1 mitigated this with a tether and a checklist, which I said at the time were weak controls. They are. The better answer is to **delete the removable part**.

The overflow becomes a moulded collar, not a loose pipe. Thermoform a 30 mm raised collar around the overflow hole in the tray floor. The collar *is* the standpipe, formed into the tray, permanently. Water above 30 mm spills over the crest and down the bulkhead. It costs nothing in tooling — the form is already being made — and it cannot be lost, left out, or fitted upside down. For the fabricated prototype trays, the equivalent is a welded-in fixed tube.

F8 is eliminated, not mitigated. Four loose standpipes come out of the BOM at ₹760.

Collar bore stays DN32 minimum. At 8 mm over the crest, DN25 passes exactly the 7.2 L/min fill rate with no margin; DN32 passes 11.9 L/min, a 1.65× margin.